

EX. NO: 2

Recover Deleted or Damaged Files from a Storage Device Using TestDisk

Date: 30-07-2026

1. Aim

To recover a deleted file from an authorized USB storage device using TestDisk and save the recovered file to a separate destination without modifying, formatting, or repairing the source device.

2. Description

TestDisk is a free and open-source data-recovery utility used to recover lost partitions, locate deleted files, and repair certain file-system structures.

TestDisk can analyze storage devices and display their partition layout. It can also list deleted files and copy selected files to another storage location.

In this experiment, a USB pendrive was used as the source device. A sample file was copied to the pendrive and then deleted. TestDisk was used to locate and recover the deleted file.

The recovered file was saved to the computer drive instead of the source pendrive. No partition-table changes or formatting operations were performed.

3. Software and Hardware Used

Requirement	Details
Operating system	Windows
Forensic tool	TestDisk 7.3-WIP
Source device	Authorized USB pendrive
Source drive	E:\
File system	NTFS
Deleted file	black-myth-wukong-7680x4320-18173
Recovery destination	D:\n\df\TestDisk-Ex02\Recovered-Files
Screenshot directory	D:\n\df\TestDisk-Ex02\Screenshots

4. Theory

When a file is deleted, the file-system entry may be removed while the file data remains on the storage device. If new data has not overwritten the previous storage area, the deleted file may be recoverable.

TestDisk provides a file-listing feature that allows deleted files to be identified. A selected file can then be copied to another storage location. For safe recovery, recovered files should not be saved to the original source device. Writing recovered data back to the source may overwrite other deleted information and reduce the possibility of further recovery. TestDisk also includes options for writing partition tables and repairing boot sectors. These options modify the source device and were not used in this experiment.

4. Procedure

Step 1: Connect the USB Pendrive

Disk Management							
File Action View Help							
Volume	Layout	Type	File System	Status	Capacity	Free Sp...	% Free
(Disk 0 partition 1)	Simple	Basic		Healthy (E...	260 MB	260 MB	100 %
(Disk 0 partition 5)	Simple	Basic		Healthy (R...	1.03 GB	1.03 GB	100 %
(Disk 0 partition 7)	Simple	Basic		Healthy (R...	260 MB	260 MB	100 %
(Disk 1 partition 1)	Simple	Basic		Healthy (E...	260 MB	260 MB	100 %
(Disk 1 partition 7)	Simple	Basic		Healthy (R...	1000 MB	1000 MB	100 %
New Volume (D:)	Simple	Basic	NTFS	Healthy (B...	250.98 GB	145.70 GB	58 %
ni (H:)	Simple	Basic	NTFS	Healthy (B...	158.20 GB	72.75 GB	46 %
nira (F:)	Simple	Basic	NTFS	Healthy (B...	98.03 GB	59.84 GB	61 %
nira_movies (G:)	Simple	Basic	NTFS	Healthy (B...	146.48 GB	103.30 GB	71 %
niranjan (E:)	Simple	Basic	NTFS	Healthy (B...	527.55 GB	377.45 GB	72 %
OS (C:)	Simple	Basic	NTFS	Healthy (B...	196.41 GB	25.77 GB	13 %
RESTORE	Simple	Basic	NTFS	Healthy (B...	28.00 GB	11.00 GB	39 %

Disk 0						
Basic 476.92 GB Online	260 MB Healthy (EFI System Par	OS (C:)		New Volume (D:)		
		196.41 GB NTFS Healthy (Boot, Page File, Crash Dump, Basic Data Parti		250.98 GB NTFS Healthy (Basic Data Partition)		
				1.03 GB Healthy (Recovery Partition)	RESTORE	260 MB
					28.00 GB NTFS Healthy (Basic Data Partition)	Healthy (R

Disk 1				
Basic 931.50 GB Online	260 MB Healthy (EFI System F	niranjan (E:)		nira (F:)
		527.55 GB NTFS Healthy (Basic Data Partition)		98.03 GB NTFS Healthy (Basic Data Partition)
			nira_movies (G:)	ni (H:)
			146.48 GB NTFS Healthy (Basic Data Partition)	158.20 GB NTFS Healthy (Basic Data Partition)

Connect the authorized USB pendrive to the Windows computer and verify that it is assigned the drive letter ****E:****. A sample file named **black-myth-wukong-7680x4320-18173** was copied to the pendrive and then deleted.

Step 2: Open TestDisk

Open the **TestDisk 7.3-WIP** folder and run **testdisk_win.exe** with administrator privileges.

At the initial screen, select **Create** to create a log file.

```

D:\n\df\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

TestDisk is free data recovery software designed to help recover lost
partitions and/or make non-booting disks bootable again when these symptoms
are caused by faulty software, certain types of viruses or human error.
It can also be used to repair some filesystem errors.

Information gathered during TestDisk use can be recorded for later
review. If you choose to create the text file, testdisk.log , it
will contain TestDisk options, technical information and various
outputs; including any folder/file names TestDisk was used to find and
list onscreen.

Use arrow keys to select, then press Enter key:
> [ Create ] Create a new log file
[ Append ] Append information to log file
[ No Log ] Don't record anything.
```

Step 3: Select the USB Device

TestDisk displays the available physical storage devices.

Identify the authorized USB pendrive using its device name and storage size, and select it.

Press **Enter** to continue.

```
D:\n\df\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

TestDisk is free software, and
comes with ABSOLUTELY NO WARRANTY.

Select a media and choose 'Proceed' using arrow keys:
Disk \\.\\PhysicalDrive0 - 512 GB / 476 GiB - SAMSUNG MZVL8512HELU-00BTW
>Disk \\.\\PhysicalDrive1 - 8056 MB / 7683 MiB - JetFlash Transcend 8GB
```

Step 4: Select the Partition Table Type

TestDisk automatically detects the partition-table type of the selected USB device.

Accept the detected option and press **Enter**.

Step 5: Analyze the Partition

Select **Analyse** from the main menu. Choose **Quick Search** to search for the partition on the USB pendrive.

```
D:\n\df\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.\\PhysicalDrive1 - 8056 MB / 7683 MiB - JetFlash Transcend 8GB
CHS 979 255 63 - sector size=512

>[ Analyse ] Analyse current partition structure and search for lost partitions
[ Advanced ] Filesystem Utils
[ Geometry ] Change disk geometry
[ Options ] Modify options
[ MBR Code ] Write TestDisk MBR code to first sector
[ Delete ] Delete all data in the partition table
[ Quit ] Return to disk selection
```

```
D:\n\df\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.\\PhysicalDrive1 - 8056 MB / 7683 MiB - CHS 979 255 63
Partition          Start      End      Size in sectors
>* FAT32            0 32 33   979 113 30   15732736 [NIRANJAN_U]
```

Step 6: List the Files

After the partition is displayed, select the required partition and press **P** to view its files and folders. Locate the deleted file: **black-myth-wukong-7680x4320-18173**

```

C:\n\df\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
* FAT32                0 32 33  979 113 30   15732736 [NIRANJAN_U]
Directory /
Previous
-rwxr-xr-x  0  0 1235020211 25-May-2022 10:52 www.1TamilMV.fans - Top Gun (1986) REMASTERED BluRay - 720p - [Tam Tel Hin Eng].mkv
drwxr-xr-x  0  0      0 8-May-2022 16:49 New folder
drwxr-xr-x  0  0    32768 25-May-2022 07:54 SCHOOL DI
-rwxr-xr-x  0  0   1774103 26-May-2022 19:17 IMG20220526191701.jpg
-rwxr-xr-x  0  0   1994538 26-May-2022 19:17 IMG20220526191724.jpg
-rwxr-xr-x  0  0   2151039 26-May-2022 19:53 IMG20220526195303.jpg
-rwxr-xr-x  0  0    799611 26-May-2022 20:32 IMG20220526203244.jpg
-rwxr-xr-x  0  0   2183111 26-May-2022 20:33 IMG20220526203342.jpg
-rwxr-xr-x  0  0   2035491 26-May-2022 20:33 IMG20220526203347.jpg
-rwxr-xr-x  0  0   2014912 26-May-2022 20:33 IMG20220526203349.jpg
-rwxr-xr-x  0  0   1901218 26-May-2022 20:34 IMG20220526203440.jpg
-rwxr-xr-x  0  0   1583015 26-May-2022 20:36 IMG20220526203654.jpg
-rwxr-xr-x  0  0   1136002 26-May-2022 20:37 IMG20220526203702.jpg
-rwxr-xr-x  0  0   2599048 26-May-2022 20:37 IMG20220526203710.jpg
-rwxr-xr-x  0  0   1463307 26-May-2022 20:37 IMG20220526203734.jpg
-rwxr-xr-x  0  0   2086514 26-May-2022 20:37 IMG20220526203735.jpg
-rwxr-xr-x  0  0   2170918 26-May-2022 20:38 IMG20220526203829.jpg
-rwxr-xr-x  0  0   3108411 26-May-2022 20:38 IMG20220526203855.jpg
-rwxr-xr-x  0  0   2419907 26-May-2022 20:39 IMG20220526203912.jpg
-rwxr-xr-x  0  0   2386743 26-May-2022 20:39 IMG20220526203912_01.jpg
-rwxr-xr-x  0  0   2018051 26-May-2022 20:39 IMG20220526203926.jpg
-rwxr-xr-x  0  0   2330464 26-May-2022 20:39 IMG20220526203931.jpg
-rwxr-xr-x  0  0   1984471 26-May-2022 20:39 IMG20220526203939.jpg
-rwxr-xr-x  0  0   1908144 26-May-2022 20:39 IMG20220526203946.jpg
-rwxr-xr-x  0  0   1552313 26-May-2022 20:40 IMG20220526204013.jpg
-rwxr-xr-x  0  0   1397439 26-May-2022 20:40 IMG20220526204023.jpg
-rwxr-xr-x  0  0   1766286 26-May-2022 20:40 IMG20220526204025.jpg
-rwxr-xr-x  0  0   1043929 26-May-2022 20:40 IMG20220526204039.jpg
-rwxr-xr-x  0  0   2296381 26-May-2022 20:40 IMG20220526204056.jpg
-rwxr-xr-x  0  0   3310741 26-May-2022 20:41 IMG20220526204100.jpg
-rwxr-xr-x  0  0   1890832 26-May-2022 20:41 IMG20220526204111.jpg
-rwxr-xr-x  0  0   2202721 26-May-2022 20:41 IMG20220526204127.jpg
-rwxr-xr-x  0  0   2202920 26-May-2022 20:41 IMG20220526204128.jpg
-rwxr-xr-x  0  0   1055411 26-May-2022 20:41 IMG20220526204137.jpg
-rwxr-xr-x  0  0   2465203 26-May-2022 20:41 IMG20220526204139.jpg
-rwxr-xr-x  0  0   1288685 26-May-2022 20:41 IMG20220526204144.jpg
-rwxr-xr-x  0  0   2599383 26-May-2022 20:41 IMG20220526204151.jpg
-rwxr-xr-x  0  0   1313436 26-May-2022 20:41 IMG20220526204156.jpg
> -rwxr-xr-x  0  0   1521020 26-May-2022 20:42 IMG20220526204231.jpg
Next
Use Right to change directory, 'h' to hide deleted files
'q' to quit, ':' to select the current file, 'a' to select all files
'C' to copy the selected files, 'c' to copy the current file

```

Step 7: Recover the Deleted File

Highlight the deleted file and press **C** to copy it.

Select the separate recovery destination:

D:\n\df\TestDisk-Ex02\Recovered-Files

Do not select the original USB pendrive as the destination.

Press **C** again to confirm the copy operation.

TestDisk should display: **Copy done! 1 ok, 0 failed**

```

-rwxr-xr-x 0 0 1408716 20-Aug-2022 00:12 KKKKK1.TXT.jpg
-rwxr-xr-x 0 0 265439304 14-Aug-2022 23:16 5_6325430905003312825.Mkv
-rwxr-xr-x 0 0 443882696 20-Oct-2022 20:23 TamilVip.tv_-_Prince_(2022)_Tamil_Proper_HQ_PreDVD_1080p_HQ_Clean_Aud_2_6GB_LQ_240p.mp4
-rwxr-xr-x 0 0 443882696 25-Oct-2022 20:42 TamilVip.tv_-_Prince_(2022)_Tamil_Proper_HQ_PreDVD_1080p_HQ_Clean_Aud_2_6GB_LQ_240p.mp4
-rwxr-xr-x 0 0 1440 13-Nov-2022 01:19 BOOTEX.LOG
-rwxr-xr-x 0 0 145129472 12-Aug-2026 20:22 _EMDUMP.MEM
>-rwxr-xr-x 0 0 8823106 30-Dec-2025 15:15 black-myth-wukong-7680x4320-18173.jpg

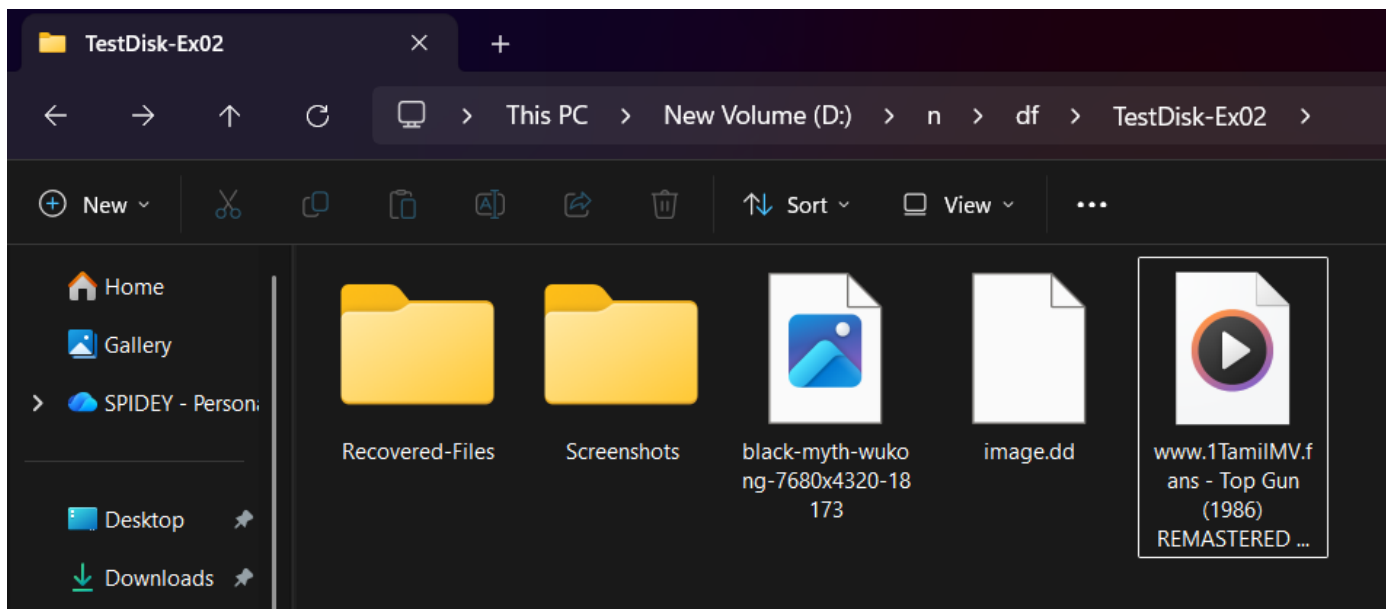
```

```

D:\n\df\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
* FAT32                0 32 33  979 113 30   15732736 [NIRANJAN_U]
Directory /black-myth-wukong-7680x4320-18173.jpg
Copy done! 1 ok, 0 failed
Previous
-rwxr-xr-x  0  0  1942331 26-May-2022 20:44 IMG20220526204421.jpg
-rwxr-xr-x  0  0  1922143 26-May-2022 20:44 IMG20220526204423.jpg
-rwxr-xr-x  0  0  1971552 26-May-2022 20:44 IMG20220526204431.jpg
-rwxr-xr-x  0  0  1969968 26-May-2022 20:44 IMG20220526204437.jpg
-rwxr-xr-x  0  0  2145170 26-May-2022 20:45 IMG20220526204453.jpg

```

Step 8: Verify the Recovered File



Exit TestDisk and open PowerShell.

Use the following command to check the recovered file:

```
Get-ChildItem "D:\n\df\TestDisk-Ex02\Recovered-Files" -Recurse
```

To view the file name, size, and modification time:

```
Get-ChildItem "D:\n\df\TestDisk-Ex02\Recovered-Files" -Recurse |
```

```
Select-Object Name, Length, LastWriteTime
```

Verify that the recovered file is present in the destination folder.

5. Observations and Results

Field	Result
TestDisk launch	Successful
Log creation	Completed
USB pendrive detection	Successful
Source drive	E:\
Quick Search	Completed
Partition listing	Displayed
Deleted file detection	Successful
File selected	black-myth-wukong-7680x4320-18173
Recovery destination	D:\n\df\TestDisk-Ex02\Recovered-Files
File copy	Successful
Recovered file verification	Successful
Source pendrive formatted	No
Source pendrive repaired	No
Partition table modified	No

7. Result

The deleted file: black-myth-wukong-7680x4320-18173 was successfully recovered from the USB pendrive assigned to: E: The recovered file was saved to: D:\n\df\TestDisk-Ex02\Recovered-Files The source pendrive was not formatted, repaired, or modified during the recovery process.

8. Conclusion

TestDisk was successfully used to analyze an authorized USB pendrive and recover a deleted file. The Analyse and Quick Search options were used to identify the partition, and the file-listing function was used to locate the deleted file. The recovered file was copied to a separate directory on the computer drive. The recovery was completed safely because no files were written back to the source pendrive and no partition-table or boot-sector changes were performed.